2021 Southwest CCDC Blue Team Packet

https://southwestccdc.com/

March 13, 2021

## Welcome

Welcome to the 2021 Southwest Regional CCDC. Congratulations on reaching the second round of the competition. This packet is your guide to rules and information for this year's competition. We update this every year so please read through this. Both before the game and during the game, if there is anything that is unclear, **please ask for clarifications**! The Regional competition is a lot more involved and intense than the qualification round. We wish everyone the best of luck!

### National CCDC Rules

This document serves as a supplement to the official National CCDC rules (http://nationalccdc.org/index.php/competition/competitors/rules). Reading this document is not a substitute for reading the National CCDC rules. All team members will be expected to have read this document and the National CCDC rules in their entirety. DO NOT JUST SKIM THEM.

## Game Organization

The game is divided into two days of competition. Between the end of day one and the start of day two the game is completely stopped. The scoring engine and all red team activity will stop for the day at the designated time, and this is enforced through firewall rules and the scoring engine. The game is divided into several teams, each named with a color.

### Gold Team

The Gold Team directs and judges the competition. A decision from Gold Team is final. Gold team is not part of the game scenario. Gold team members are to be inherently trusted.

- Andrew Kongs — Competition Director
- Gavin Bauer — Assistant Competition Director

### Black Team

As the network and game operations team, members of the Black Team ensure that the network and systems are functioning as intended.

- Brady Deetz — Black Team Co-Lead
- George Louthan — Black Team Co-Lead
- Peter Gehres — Build Lead
- Aaron Glenn
- Riley Childs
- Bradley D'Amato

### White Team

White Team operates the business aspects of the competition. They provide competition support and manage the game scenario, grade injects, and perform other important competition tasks. Trust the white team.

- Taylor Smith — White Team Lead

### Red Team

Your adversary. Members from the Red Team will attempt to bring your services down, infiltrate the company, and exfiltrate your data. The red team is composed of industry security consultants and professionals. Unlike an attacker in a live environment, Red Team is here to help you learn and serve as your training partner, providing feedback.

### Blue Team

That's you, the student competitors. You are tasked with operating and securing the network and performing business operations within the scenario.

## Event Schedule

### Friday, March 19th

- 9:00 A.M. – Competition Briefing, Discord
- 10:00 A.M. – Game Start
- 6:00 P.M. – Game Stop
- 6:00 P.M. – Day 1 Debrief, Discord

### Saturday, March 20th

- 9:30 A.M. – Competition Briefing, Discord
- 10:00 A.M. – Game Start
- 6:00 P.M. – Game Stop
- 6:00 P.M. – Day 2 Debrief, Discord

### Sunday, March 21st

- 9:30 A.M. – Sunday Briefing, Discord
- 10:00 A.M. – Individual Team Debriefs, Discord
- 2:30 P.M. – Career Fair, Zoom
- 4:30 P.M. – Sponsor Keynotes, Webex
- 5:30 P.M. – Closing Ceremony and Awards, Webex

## Business Scenario — Blipper

Blipper is a social media platform that allows people to share short 140 character posts, along with a photo, gif, or meme. Blipper has just acquired upstart Florp to add video capabilities to its platform. Dave "Blip" Blipananopoulos founded Blipper as part of a drunken wager at DEFCON. His arch-nemesis from college, Steve Trisolian, bet Dave that he couldn't build a social network and gain 10,000,000 users in a month. Blipper achieved milestone in 27 days and went to become dominant in the field of microblogging.

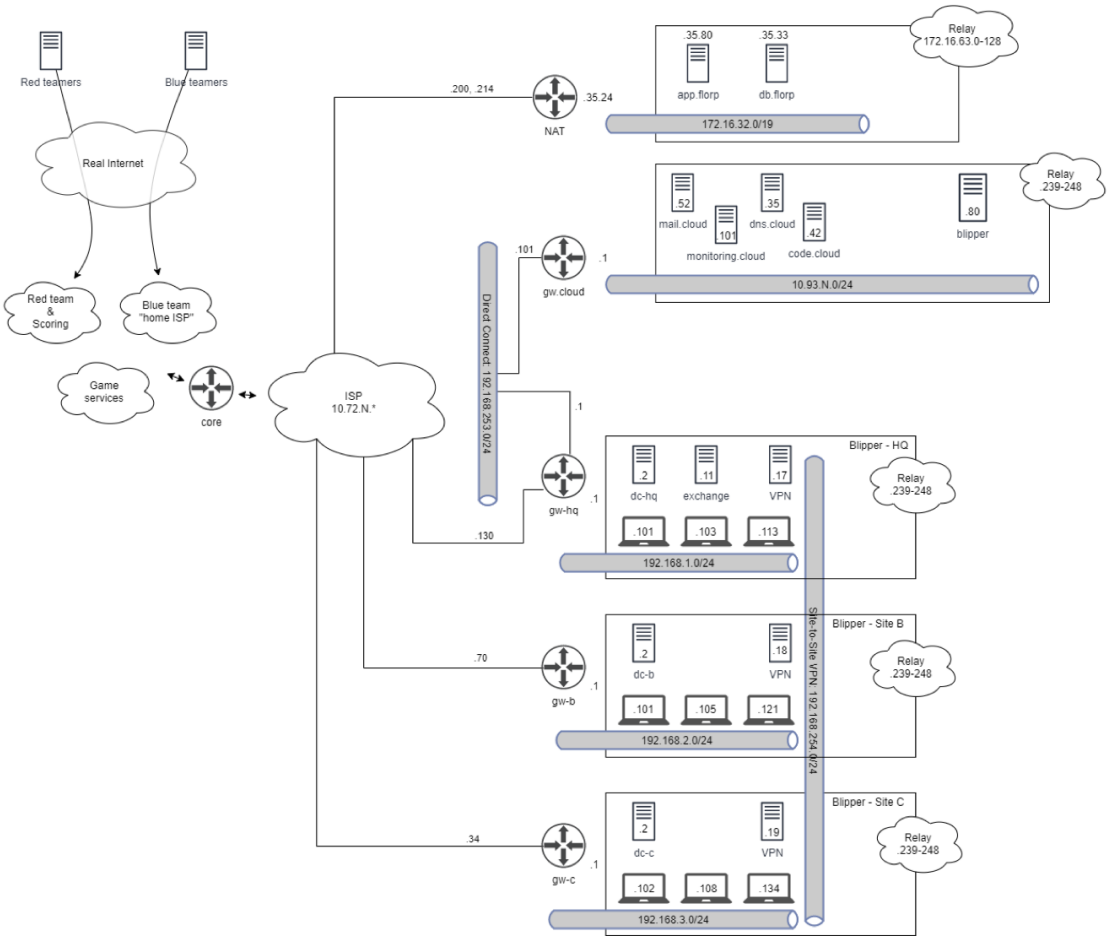
### Team portal

For injects, policy, incident responses (IR), and scoring, teams will access the portal at https://portal.southwestccdc.com. Your team's credentials for the portal are created by your Team Captain. Each team shares a single set of credentials for the portal. The portal and the OpenNebula accounts are separate.
The portal also shows your team's service up/down status and a manual with details how to use various features. The manual can also be accessed directly at https://portal.southwestccdc.com/manual.

### Gamestationvm and VPN

https://1drv.ms/u/s!AoFtBT9cDltHgZsgYdMeU_9O26lGtA?e=ztyNzV
The link above contains configuration files required to connect to the competition network. It also contains a virtual image called the gamestationvm, which is preconfigured to connect to the VPN. You may use the preconfigured VM, or you may connect from your own computer, or both.
Importantly, if you choose to use your own computer to connect directly via the VPN, you may have some extra exposure. We recommend you use the gamestationvm as a sandbox.

### Blipper Network Diagram




## CCDC Special Edition: Work From Home

Remote work, especially this year, is a norm. Blue teams work for a company with multiple distributed offices—and you are not located at any of them.
You will not be working from a simulated office; instead, you will be working from a simulated home ISP.

## Discord

Students will use Discord to communicate among their team members as well as with White, Gold, and Black teams during the competition. This discord server is out of scope for the red team, so you can trust it.

Teams must provide a roster of their numeric Discord IDs **no later than Wednesday, March 17th** so we can assign them to the correct channels. All black, gold, and white team personnel will also be available on discord. Both dedicated help (questions only relevant to your team) and general help channels are available on the SWCCDC discord.
Some injects will be scored using discord voice, video, or text channels. If an inject indicates that it will be scored via Discord or by presentation, a dedicated scoring channel will be used for each team.
**Do not send Discord DMs to SWCCDC staff or volunteers.** Do not submit inject responses or incident reports on Discord.

## Etherpad

The competition environment includes Etherpad, a shared text and document editor for the Blue Teams. It is out of scope for Red Team activity and can be safely used for any collaborative text or document editing. Access it at `https://etherpad.quals.southwestccdc.com`. We suggest you click the big green "New Pad" button to create a document, and share the URL with your teammates, rather than choosing an easy to guess document name. NOTE: Do not submit links to Etherpad as inject submissions! It supports text, markdown, and PDF export, so use those and then upload to the portal!

## Business and Technical Injects

Teams are responsible for answering requests, memos, and correspondence in a professional manner. Take care to understand what each request is asking for in detail. Some injects request a technical task, others request documentation of some kind. Do not provide a short memo when asked for a report and do not provide a report as a paragraph. Responses to injects are scored heavily and can influence the final outcome of the game.
Responses are penalized heavily once they are even one minute past-due. A memo reply requested by 10:00 A.M. that is received at 10:01 will be penalized by 50% before any subject-matter scoring is completed.
Inject responses must be submitted through the portal to be scored.
There may be clarifications, conversations, meetings, video chats, or other inject-related communications on Discord. All of these are welcome, but the graded activity is submitted to the portal unless otherwise indicated in the inject text. Upload your entire inject response to the portal. **Your response should be uploaded. Do not submit a link.**
DO use your assigned team number in your inject responses. DON'T use your school name.
Inject graders only review the last submitted response of an inject. Do not reference previously uploaded responses.

Additionally, all injects for Day 1 must be submitted before the end of Day 1, even if they are late within the day. No points will be awarded for Day 1 injects submitted on Day 2. Similarly, all injects for Day 2 are due before the end of the game on Day 2.

## Services

Services are checked on random intervals every few minutes. Each of these service checks are worth points. If a service is down, it will not grant any points to the team. All services will be functional when you gain access to the network. The Portal shows the status of each service at its most recent check. It is the team's responsibility to secure these services and ensure their continued operation. You will be provided a list of services you are operating along with the network diagram a few days before the competition.
Services must remain on the same major version of the software operating the service, same major OS version, and the same machine for the first day of the competition, **unless otherwise indicated by an inject**. For example, a machine running Windows 2008 R2 Server and IIS 7.x must continue to run that version combination for Day 1. On the second day of the competition, requests for modernization can be sent to the CEO/CIO for consideration. Submit these requests through the portal's Host Change tab.

## Scoring

Teams accumulate points by keeping network services functioning and by completing injects. Injects and Services each account for approximately half of the total points available to teams. Red Team activity deducts points accumulated from injects and services scores.
**NOTE**: Due to the configuration of the remote environment this year, we ask that you do not change the IP of scored services. Changing the DNS record for the host will not change which IP the scoring engine checks. If you change the IP of a scored service, you will not receive points for that service.

## Red Team

Red Team engagement will occur as soon as the game begins. Red Team will document all attacks against your network, and you have the opportunity to submit an incident response memo whenever you have been compromised. The portal is out of scope for Red Team.
**NOTE**: Red team has been advised not to pursue "scorched-earth" style attacks. If your machine doesn't boot and the partition table is gone, it was likely a Blue Team action.

**Due to the remote nature of the game for 2021**, Blue Teams will have no opportunities for direct interaction with anyone that might be Red Team. Red Team will not have access to the main Discord instance nor will not have a way to provide spurious injects.

## Incident Response

Red Team incidents can range from being complex and involving multiple steps to simple "smash-and-grab" exploits. You may submit a incident response for every "Attack" that has a material impact. An "attack" is any significant step in a compromise chain. Include a detailed description of what machine you believe was compromised, its IP address, the timeframe you believe the attack occurred, and what can be done to mitigate future similar attacks. A memo that reads "someone scanned us" will be disregarded. Accurate incident responses can recover up to half of the points lost from the incident. A few examples:

1. If the Red Team uses an exploit to achieve unauthenticated RCE on a web-server and then changes the configuration of the web-server to block requests, that would be a single incident response.

2. If the Red Team is able to download a list of admin accounts from a poorly configured FTP server, then uses those credentials to login to an admin interface and disable a service, that is two incident responses. The first report is a credential leak from the FTP machine; the second is using access to disable the service.

3. If the Red Team social engineers a team member to get admin credentials, then uses those to install a web-shell on a web-server, then uses that web-shell to install persistence, then uses the persistence shell to exfiltrate credit card data from customers, that would be three responses. First would be the social engineering credential leak, second would be the installation of the web-shell and persistence, and third would be the exfiltration of credit card data.

4. If the Red Team port-scans your network, that would not be a report.

5. If the Red Team browses to an unsecured admin interface and does not change anything, that would not be a report.

6. If you discover a list of accounts on an FTP server but have no proof anyone accessed that list, that would not be a report.

## In-Game Money

Teams will be provided a "budget" to buy things they find useful like in-game consulting. Initially, each team will be provided $5000, available immediately at the

start of the game. Large injects and game events may provide your team access to more money to spend, however you should not count on this. New items may be made available for purchase throughout the game. Teams may view their current balance and spend their budget on available items through the "Store" tab of the Portal.

## Gold / Black team support

Using Discord, you will be able to reach a White Team member who can direct your question to the correct person. If you believe something is severely wrong with the competition please request an escalation to Gold or Black Team.
If you unnecessarily or repeatedly call for Gold/Black Team support and the issue is determined to be Blue or Red Team's fault, the Blue Team may be penalized up to $500 from their budget, at the discretion of Gold/Black Team. This effectively becomes "purchased consulting." For example, if a laptop's Ethernet port stops working, that is not Blue Team's fault and there will be no penalty. If the Red Team breaks into a router under Blue Team control and prevents traffic from being routed, that is Blue/Red Team's fault and the Blue Team will be penalized. The penalties only apply for teams who call for Gold/Black support. If you solve the issue on your own, you do not risk the penalty.
However, if you believe something is wrong, reach out to Black or Gold Team. This applies to scoring as well. If you believe a service is not being scored correctly, the same penalty applies if scoring is found to be working correctly. The purpose of this is to prevent unnecessary escalations and overloading Gold/Black Team with "tech support".

## Post-Game

The **winner** of the Southwest Regional CCDC will advance to the National CCDC competition in April. The **team who places second**, will proceed on to the National CCDC "wildcard" round. The second-place teams from every region will then compete, with the winner of the wildcard round then advancing to National CCDC. Teams in first, second and third place will be awarded trophies for their achievement. Even if you don't win, we hope the competition will be an exhilarating learning experience.

## About the Southwest CCDC Team

Southwest CCDC is operated entirely by a dedicated group of volunteers composed of CCDC veterans, industry professionals, and academia. This team spends many months of our own time to design, build, and operate the game. Southwest CCDC is an event run by TALON Cyber League, an independent 501(c)(3) non-profit whose mission is to prepare students for careers in cybersecurity by engaging them with competitive, experiential learning. Much of the physical infrastructure and resources for our

on-premises activities are provided by the The University of Tulsa, who graciously hosts our equipment and provides the venue for the regional game.

## Resource Links

- Full Game Rules @ National CCDC
- Mubix's How to win CCDC GitHub
- National CCDC Team Prep Guide
- SWCCDC — "What is CCDC video" on YouTube (15 min)
- Intro to CCDC and 2018 scenario review video on YouTube
- Intro to CCDC and 2018 scenario review slides (PDF)
- Tim's Twitter Thread from SWCCDC 2018

## Books

- Blue Team Field Manual (BTFM) by Alan White and Ben Clark
- The Practice of System and Network Administration, by Thomas A. Limoncelli, Christina J. Hogan, and Strata R. Chalup
- Learn Windows Powershell in a Month of Lunches, by Donald W. Jones and Jeffrew Hicks
- TCP/IP Guide by Charles M. Kozierok (No Starch Press)
- The Phoenix Project